



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Smart Travel Route Planning and Budget Optimization Platform Using Java

[CSA0908 – PROGRAMMING IN JAVA]

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Abstract



- The platform helps users plan trips by combining route selection, transportation choices, accommodation, food and activities.
- Users enter source, destination, travelers, trip duration, dates, budget and travel preferences.
- The system compares transport and trip combinations and generates Luxury, Balanced and Budget Saver plans.
- A day-by-day itinerary is generated with morning, lunch, afternoon, evening and night activities.
- User accounts allow trips to be saved, viewed and modified later.

Introduction

- Planning a trip manually requires checking multiple transport, hotel, food and activity options.
- Different choices make it difficult to know whether the complete trip will stay within the available budget.
- Users may spend extra time comparing routes, prices, ratings and daily activities.
- The proposed Java-based website combines these tasks in one platform.
- It aims to provide a practical, personalized and budget-aware travel planning experience.

Problem Identification and Analysis

- Travel options are scattered across different sources and are difficult to compare quickly.
- Users may select transport or hotels without considering the total trip cost.
- A fixed budget must be divided across travel, accommodation, food and activities.
- Creating a practical day-by-day itinerary manually can be time-consuming.
- A smart system is needed to compare alternatives and produce plans that match user preferences and budget.

Solution Development Process

- Identify requirements such as route search, budget limits, trip preferences, itinerary and user accounts.
- Design the system workflow, database structure and interaction between the web interface and Java backend.
- Develop the application using Java and Spring Boot with REST-based services.
- Store users, destinations, transport options, hotels, activities and saved trips in MySQL.
- Implement the budget optimization logic and itinerary generation.
- Test route calculation, cost calculation, plan generation, authentication and trip-history functions.



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Solution Design

Tools and Technologies Used

- Java and Spring Boot are used for the main backend services.
- Spring MVC and REST APIs connect the website with application services.
- Spring Data JPA is used for database access and entity management.
- MySQL stores users, trips, destinations, transport, hotels, activities and costs.
- HTML, CSS, JavaScript and Bootstrap provide the website interface.
- Route, weather and places APIs can supply external travel information.

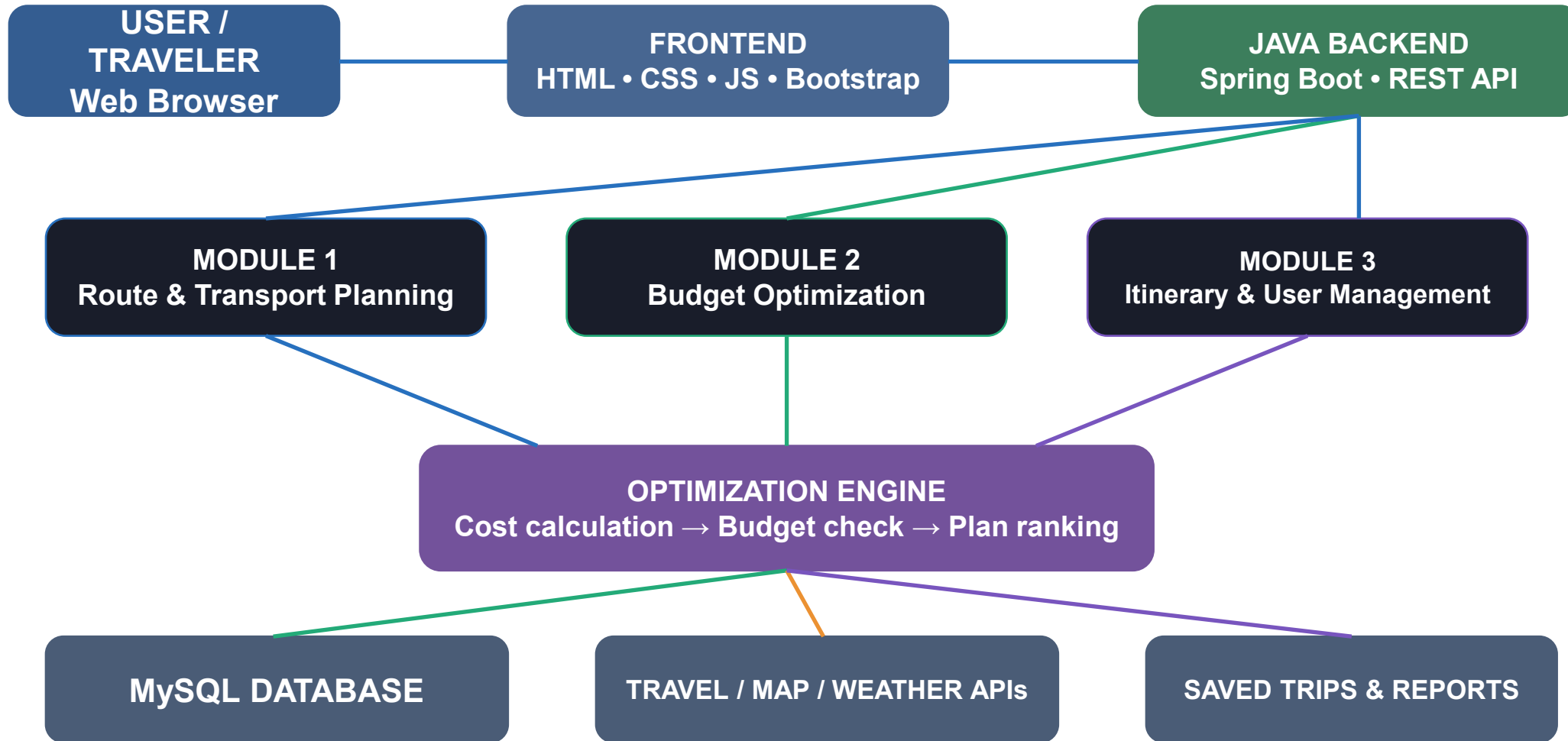
Hardware Requirements

- Computer/laptop with basic configuration.
- Minimum 4 GB RAM and sufficient storage.

Software Requirements

- Windows, Linux or macOS.
- JDK, IDE, MySQL and modern web browser.

System Architecture





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Module 1: Route & Transport Planning

Route Planning

The module identifies practical travel options from the source to the destination.

After entering trip details, the system can:

- Compare Flight, Train, Bus and Cab options
- Calculate distance and estimated travel time
- Estimate ticket cost for all travelers
- Rank alternatives by cost, time and preference

Transport Comparison

The module receives route information from stored data or external APIs.

Example: Chennai → Goa | 3 travelers

- Flight – Faster, higher cost
- Train – Balanced travel time and cost
- Bus – Lower cost, longer travel time
- Cab – Flexible, but may cost more

Module-1 Source Code

The screenshot displays an IDE interface with the following components:

- EXPLORER:** Shows the project structure for `JAVA_CAP`. The `src` directory contains `main`, which includes `java\com\travelplanner`. Under `java\com\travelplanner`, there are subdirectories `controller`, `dto`, `model`, `repository`, and `service`. The `service` directory contains several Java files, with `RouteService.java` selected.
- Editor:** Displays the source code for `RouteService.java`. The code is as follows:

```
src > main > java > com > travelplanner > service > RouteService.java
2
3 import com.travelplanner.dto.TransportOptionDto;
4 import com.travelplanner.model.Destination;
5 import com.travelplanner.model.TransportRoute;
6 import com.travelplanner.repository.DestinationRepository;
7 import com.travelplanner.repository.TransportRouteRepository;
8 import org.springframework.stereotype.Service;
9
10 import java.util.*;
11
12 @Service
13 public class RouteService {
14
15     private final TransportRouteRepository transportRouteRepository;
16     private final DestinationRepository destinationRepository;
17
18     // City Geolocation lookup for distance calculation
19     private static final Map<String, double[]> CITY_COORDS = new HashMap<>();
20
21     static {
22         // [Latitude, Longitude] - India Cities
23         CITY_COORDS.put("CHENNAI", new double[]{13.0827, 80.2707});
24         CITY_COORDS.put("GOA", new double[]{15.2993, 74.1240});
25         CITY_COORDS.put("MUMBAI", new double[]{19.0760, 72.8777});
26         CITY_COORDS.put("DELHI", new double[]{28.6139, 77.2090});
27         CITY_COORDS.put("BENGALURU", new double[]{12.9716, 77.5946});
28         CITY_COORDS.put("BANGALORE", new double[]{12.9716, 77.5946});
29         CITY_COORDS.put("HYDERABAD", new double[]{17.3850, 78.4867});
30         CITY_COORDS.put("KOLKATA", new double[]{22.5726, 88.3639});
31         CITY_COORDS.put("PUNE", new double[]{18.5204, 73.8567});
32         CITY_COORDS.put("JAIPUR", new double[]{26.9124, 75.7873});
33         CITY_COORDS.put("MANALI", new double[]{32.2396, 77.1887});
34         CITY_COORDS.put("MUNNAR", new double[]{10.0889, 77.0595});
35         CITY_COORDS.put("UDAIPUR", new double[]{24.5854, 73.7125});
36         CITY_COORDS.put("VARANASI", new double[]{25.3176, 82.9730});
```
- Bottom Bar:** Shows the current file is `main`, with 0 errors and 0 warnings. The status bar indicates the file is `charan505 (1 week ago)`, `Ln 10, Col 20`, `Spaces: 4`, `UTF-8`, `LF`, `{ }` Java, and includes links for `Sign In`, `Go Live`, and `Prettier`.

MODULE-1 OUTPUT-1



WanderWise
BUDGET & ROUTE ENGINE

Plan Trip

Explore

Architecture

My Trips **0**

Sign In / Register



Trip Parameters & Preferences

Customize your origin, target budget, and vibe for algorithm optimization

Starting Location (Origin)

Chennai



Destination

Goa (Beach)



Number of Travelers

−

3

+

Trip Duration (Days)

−

5

+

Maximum Total Budget (₹)

₹18,000

₹30,000

₹50,000

₹85,000

₹ 30000

Travel Vibes & Preferences (Select All That Apply)



Beaches & Coast



Sightseeing & Forts



Nightlife & Clubs



Water Sports & Treks



Foodie & Local Eateries



Spa & Leisure



Generate Optimized Travel Plan

MODULE-1 OUTPUT-2

ORIGIN

Chennai

DESTINATION

Goa

884 km

3 Travelers

5 Days

Budget: ₹ 30,000

Multi-Modal Transport Options Evaluated:

Express Train
(Sleeper Class)

BUDGET

Operator: Vasco Da Gama Express (Sleeper)
Travel Time: ~18 hrs (960 km)
Departure: 03:30 PM

₹495 / person
Group Total: ₹1,485 34 kg CO₂

Superfast
Express (AC 3-Tier)

BALANCED

Operator: Vasco Da Gama Express (3AC)
Travel Time: ~17.5 hrs (960 km)
Departure: 03:30 PM

₹1,380 / person
Group Total: ₹4,140 38 kg CO₂

Volvo AC Multi-Axle Sleeper

BALANCED

Operator: SRS Volvo Multi-Axle AC Sleeper
Travel Time: ~16 hrs (915 km)
Departure: 06:00 PM

₹1,450 / person
Group Total: ₹4,350 55 kg CO₂

Direct Flight

LUXURY

Operator: IndiGo 6E-651
Travel Time: ~1.8 hrs (910 km)
Departure: 08:15 AM

₹4,200 / person
Group Total: ₹12,600 125 kg CO₂



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Module 2: Budget Optimization



- User enters the maximum budget for the complete trip.
- The system divides the budget across transportation, hotel, food and activities.
- Different combinations are evaluated using total cost and preference scores.
- Plans that exceed the budget are filtered or adjusted using lower-cost alternatives.
- Three final options are generated: Luxury, Balanced and Budget Saver.

Module-2 Source Code

The screenshot shows an IDE with the following components:

- EXPLORER:** Displays the project structure. The `src > main > java > com > travelplanner > service` path is selected, showing `BudgetOptimizationService.java` and other service files.
- Editor:** Shows the source code for `BudgetOptimizationService.java`. The code is as follows:

```
1 package com.travelplanner.service;
2
3 import com.travelplanner.dto.*;
4 import com.travelplanner.model.Destination;
5 import com.travelplanner.model.Hotel;
6 import com.travelplanner.repository.DestinationRepository;
7 import com.travelplanner.repository.HotelRepository;
8 import org.springframework.stereotype.Service;
9
10 import java.util.*;
11
12 @Service
13 public class BudgetOptimizationService {
14
15     private final DestinationRepository destinationRepository;
16     private final HotelRepository hotelRepository;
17     private final RouteService routeService;
18     private final ItineraryService itineraryService;
19
20     public BudgetOptimizationService(DestinationRepository destinationRepository,
21                                     HotelRepository hotelRepository,
22                                     RouteService routeService,
23                                     ItineraryService itineraryService) {
24         this.destinationRepository = destinationRepository;
25         this.hotelRepository = hotelRepository;
26         this.routeService = routeService;
27         this.itineraryService = itineraryService;
28     }
29
30     /**
31      * Core Algorithm: Multi-Objective Constrained Optimization & Tier Generation
32      */
33     public OptimizationResponse optimizeTrip(TripRequest request) {
34         OptimizationResponse response = new OptimizationResponse();
```
- Bottom Bar:** Shows the current file is `main`, with 0 errors and 0 warnings. The status bar also indicates `Ln 1, Col 1`, `Spaces: 4`, `UTF-8`, `LF`, `{ }`, `Java`, and provides links for `Sign In`, `Go Live`, and `Prettier`.

MODULE 2 : OUTPUT-1

**WanderWise**
BUDGET & ROUTE ENGINE

[Plan Trip](#)[Explore](#)[Architecture](#)[My Trips 1](#)

**karthik mogill**
Sign Out

Choose Your Tailored Optimization Tier

Compare luxury convenience vs balanced smart-value vs strict budget minimization

COMFORT TIER

Plan A: Luxury & Comfort
Fast flights / 1st AC, 5★ resorts, fine dining

₹1,62,435
₹54,145 / person
~~₹2,07,917~~ **Save ₹45,482**

 **Transport:** Direct Flight (Round-trip)

 **Stay:** 5★ Taj Exotica Resort & Spa (Sea View Deluxe Villa)

 **Dining:** ₹34,500 total allocation

 **Activities:** ₹27,000 entry passes

 **Buffer:** ₹7,735 safety reserve

Select Comfort Plan

RECOMMENDED ★

Plan B: Balanced Smart-Value
AC 3-tier / deal flights, boutique stays, top attractions

₹66,728
₹22,243 / person
~~₹83,410~~ **Save ₹16,682 (22.8%)**

 **Transport:** Volvo AC Multi-Axle Sleeper (Round-trip)

 **Stay:** 4★ Boutique Coastal Palms Resort (Deluxe Pool View Room)

 **Dining:** ₹16,500 total allocation

 **Activities:** ₹12,750 entry passes

 **Buffer:** ₹3,178 safety reserve

Selected (Recommended)

BUDGET SAVER ★

Plan C: Budget Saver
Sleeper train/bus, top homestay, street food & free sights

₹25,746
₹8,582 / person
~~₹31,410~~ **Save ₹5,664**

 **Transport:** Express Train (Sleeper Class) (Round-trip)

 **Stay:** 3★ Zostel / Backpacker Beach Haven (Private Double / Dorm Room)

 **Dining:** ₹7,500 total allocation

 **Activities:** ₹5,250 entry passes

 **Buffer:** ₹1,226 safety reserve

Select Budget Saver

MODULE 2:OUTPUT-2



WanderWise
BUDGET & ROUTE ENGINE

Plan Trip

Explore

Architecture

My Trips **1**



karthik mogili
Sign Out

Categorized Spending Matrix

Plan B: Balanced
(Recommended)



- Transport (13%) **₹8,700**
- Stay & Hotel (38.4%) **₹25,600**
- Food & Dining (24.7%) **₹16,500**
- Activities (19.1%) **₹12,750**
- Reserve (4.8%) **₹3,178**

Live Category Budget Re-balancer

Instant Recalculation

Tweak any category below to immediately see how the total package and emergency buffer dynamically adjust:

Accommodation Quality

Budget Homestay (₹1.2k/n)

3-4★ Boutique Resort (₹3.2k/n)

5★ Luxury Spa (₹8.5k/n)

Dining & Culinary Experience

Authentic Street Cafes

Popular Local Restros

Rooftop & Fine Dining

Activity & Sightseeing Pace

Relaxed (1 paid/day)

Standard (2-3 spots/day)

High Octane (Water Sports/VIP)

Adjusted Total Cost:

₹66,728

Budget Status:

▲ Exceeds by ₹36,728



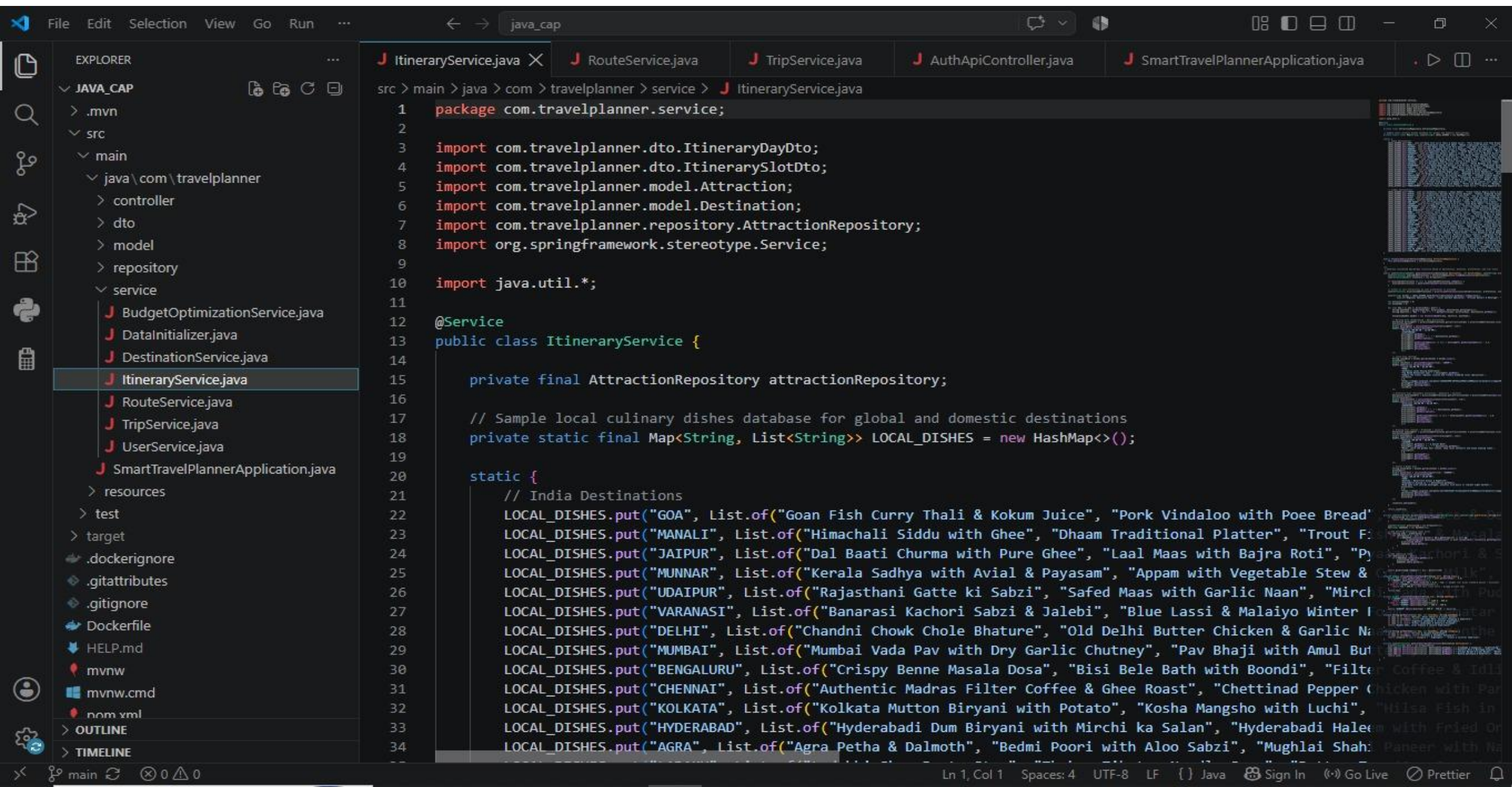
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Module 3: Daily Itinerary & User Management



- Generates a day-by-day itinerary based on trip duration, destination and preferences.
- Each day can include morning, lunch, afternoon, evening and night activities.
- Local food recommendations can be added according to budget and preferences.
- Users can register, log in, save trips and view previous trip history.
- Saved itineraries can be modified later without creating the trip again.

Module-3 Source Code



MODULE 3: OUTPUT

WanderWise
BUDGET & ROUTE ENGINE

[Plan Trip](#)[Explore](#)[Architecture](#)[My Trips](#)

[Sign In / Register](#)

Goa

Leaflet | © CartoDB, OpenStreetMap

Optimized Day-by-Day Itinerary

Chronological morning-to-night timeline with localized dishes and entry fees

[Print / Export Itinerary](#)[Save This Plan](#)

Day 1

Day 2

Day 3

Day 4

Day 5

Day 1: Arrival, Hotel Check-in & Iconic Sunset

Theme: Arrival & Leisure Exploration | Estimated Day Cost: ₹1,200

Morning (09:00 AM - 12:30 PM)

SIGHTSEEING

Baga & Colangute Beach

Baga & Colangute Beach, Goa

Vibrant beaches famous for water sports, beach shacks, and sunbathing.

Free Entry

~2 hrs

Lunch (01:00 PM - 02:30 PM)

FOOD

Authentic Local Dining Experience

Top Rated Restaurant near Baga & Colangute Beach

Sample the iconic regional cuisine and freshly prepared local specialties.

Local Dish to Try: Goan Fish Curry Thal & Kokum Juice

₹450

~1.5 hrs

Afternoon (03:00 PM - 05:30 PM)

ADVENTURE

Historic Fort Aguada & Lighthouse

Historic Fort Aguada & Lighthouse, Goa

17th-century Portuguese fortress offering 360-degree Arabian Sea views.

₹100

~3.5 hrs

Evening (06:00 PM - 08:00 PM)

LEISURE

Anjuna Flea Market & Sunset & Sunset Walk

Anjuna Flea Market & Sunset, Goa

Relax, catch the golden hour sunset, shop local souvenirs and enjoy evening vibes.

Free Entry

~2 hrs

Night (08:30 PM - 10:30 PM)

FOOD

Rooftop / Waterfront Dinner & Nightlife

Atmospheric Dining in Goa

Wind down with chilled beverages, acoustic live music or vibrant night markets.

Local Dish to Try: Pork Vindaloo with Paoe Bread

₹650

~2 hrs

Results and Discussion

- The proposed platform combines route planning, budget analysis and itinerary generation in one workflow.
- Users can compare transport choices using distance, time and estimated cost.
- The optimization module can present multiple plans instead of a single fixed recommendation.
- The itinerary module reduces the effort required to organize activities and meals day by day.
- User accounts and trip history provide continuity for future travel planning.

Future Scope

- Real-time price and availability integration for flights, trains, buses and hotels.
- GPS and live-map support for more accurate route and distance planning.
- Weather-aware itinerary adjustment and activity recommendations.
- AI-based personalized recommendations using previous trips and user preferences.
- Mobile application support with notifications and offline access.
- Dynamic optimization when prices, weather or travel conditions change.

Conclusion

- The proposed Java platform provides a single website for route planning, cost comparison and itinerary generation.
- Its main optimization feature converts a user's budget into practical travel plans with different comfort levels.
- The three-module structure demonstrates Java backend development, database connectivity, REST services and application logic.
- The platform can be expanded with real-time APIs, weather data and intelligent recommendations.
- Overall, the project provides a practical and scalable Java capstone solution for smart travel planning.



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THANK YOU